

Elevon Direct-Drive Pin Load Calculation - Revised Safety Factor

Updated preliminary calculation using SF = 1.2 and SF = 1.5 maximum

1. Purpose and update

This document updates the previous elevon direct-drive pin calculation by reducing the mechanical safety factor. The aerodynamic loads are unchanged. The revised design torque is shown for SF=1.2 and SF=1.5, with SF=1.5 used for the pin-sizing conclusion.

The single elevon span is assumed to be one V-tail half-panel because the design table states that the elevon spans almost the whole V-tail surface. Therefore $b_e = 0.495$ m. If the final elevon span changes, the aerodynamic force and hinge moment scale approximately linearly with b_e .

For the direct-drive pin concept, the pin torque is approximately the elevon hinge moment. There is no servo arm / aileron horn ratio. The design torque is $T_{\text{design}} = SF * HM_e / \eta$.

Table 1. Input values and assumptions.

Property	Symbol	Value	Unit	Comment
Air density	ρ	1.17	kg/m ³	From aircraft sizing table
Sizing speed	V	24.0	m/s	Use Vmax if higher
Stall speed	V_stall	14.0	m/s	Comparison only
Kinematic viscosity	ν	1.44e-05	m ² /s	From aircraft sizing table
Wing mean chord	c_w	0.300	m	Used for Reynolds check
V-tail angle to horizon	theta	37.9	deg	From screenshot
Tail moment arm	L_tail	0.90	m	HT/VT moment arm from screenshot
Total V-tail area	S_V-tail	0.2452	m ²	From screenshot
Full V-tail span	b_V-tail	0.990	m	From screenshot
V-tail chord	c_V-tail	0.248	m	From screenshot
Single elevon chord	c_e	0.099	m	From screenshot
Single elevon span	b_e	0.495	m	Assumed one elevon covers one V-tail half span
Center-of-pressure arm	x_cp	0.0396	m	0.40*c_e behind hinge
Safety factor range	SF	1.2 and 1.5	-	Revised from previous SF=3
Drive efficiency	η	0.90	-	Direct pin/coupling efficiency estimate
Pin length	L_pin	30.0	mm	User-proposed total pin length

2. Aerodynamic quantities

$$q = 0.5 \rho V^2 = 0.5 \times 1.17 \times 24^2 = 336.96 \text{ Pa}$$

$$q_{\text{stall}} = 0.5 \rho V_{\text{stall}}^2 = 114.66 \text{ Pa}$$

$$q / q_{\text{stall}} = 2.94$$

$$Re = V c_w / \nu = 24 \times 0.30 / 1.44e-05 = 500000$$

$$A_e = b_e c_e = 0.495 \times 0.099 = 0.04900 \text{ m}^2$$

$$x_{\text{cp}} = 0.40 c_e = 0.0396 \text{ m}$$

If a different maximum speed is later used, scale all aerodynamic forces, hinge moments, direct-drive torques, and pin loads by $(V_{\max}/24)^2$.

3. Elevon normal force

$C_{N,e} = 2 \pi \tau \delta_e$, with δ_e in radians

$F_{N,e} = q A_e C_{N,e}$

Table 2. Estimated aerodynamic normal force on one elevon at 24 m/s.

Deflection	$C_{N,e}$ lower	$C_{N,e}$ conservative	Force lower	Force conservative
10 deg	0.548	1.097	9.05 N	18.11 N
15 deg	0.822	1.645	13.58 N	27.16 N
20 deg	1.097	2.193	18.11 N	36.22 N

4. Hinge moment and revised direct-drive torque

$HM_e = c_h q A_e c_e$

$HM_e = F_{N,e} x_{cp}$, with $x_{cp} = 0.0396$ m

$T_{\text{design}} = SF \times HM_e / \eta$

Table 3. Hinge moment and direct-drive design torque for one elevon.

Deflection	HM lower	HM conservative	T_{design} , SF=1.2	T_{design} , SF=1.5
10 deg	0.359 Nm	0.717 Nm	0.956 Nm / 9.7 kg cm	1.195 Nm / 12.2 kg cm
15 deg	0.538 Nm	1.076 Nm	1.434 Nm / 14.6 kg cm	1.793 Nm / 18.3 kg cm
20 deg	0.717 Nm	1.434 Nm	1.912 Nm / 19.5 kg cm	2.390 Nm / 24.4 kg cm

5. V-tail force components and pitch/yaw moments

$F_{\text{vertical}} = F_{N,e} \cos(\theta)$, $F_{\text{side}} = F_{N,e} \sin(\theta)$

$\theta = 37.9$ deg, $\cos(\theta) = 0.789$, $\sin(\theta) = 0.614$

Table 4. Approximate aircraft-level V-tail moment components from conservative elevon force.

Deflection	Vertical component one side	Side component one side	Symmetric pitch moment	Differential yaw moment
10 deg	14.29 N	11.12 N	25.72 Nm	20.02 Nm
15 deg	21.43 N	16.69 N	38.58 Nm	30.03 Nm
20 deg	28.58 N	22.25 N	51.44 Nm	40.05 Nm

6. Revised pin torsion and twist

Worst revised pin torque uses 20 deg conservative case with SF=1.5: $T = 2.390 \text{ Nm}$

$$\tau_{\max} = 16 T / (\pi d^3)$$

$$d_{\min} = [16 T / (\pi \tau_{\text{allow}})]^{1/3}$$

Table 5. Minimum solid pin diameter for the revised SF=1.5 torque.

Allowable shear stress	Required solid pin diameter
50 MPa	6.24 mm
100 MPa	4.96 mm
150 MPa	4.33 mm
200 MPa	3.93 mm

$$J = \pi d^4 / 32$$

$$\theta = T L / (J G), \text{ with } L = 30 \text{ mm}$$

Table 6. Pin torsional stress and twist under the revised SF=1.5 torque.

Pin diameter	Torsional shear stress	Twist, aluminium $G=26 \text{ GPa}$	Twist, steel $G=79 \text{ GPa}$
3 mm	450.9 MPa	19.87 deg	6.54 deg
4 mm	190.2 MPa	6.29 deg	2.07 deg
5 mm	97.4 MPa	2.58 deg	0.85 deg
6 mm	56.4 MPa	1.24 deg	0.41 deg
7 mm	35.5 MPa	0.67 deg	0.22 deg
8 mm	23.8 MPa	0.39 deg	0.13 deg

7. Bearing pressure check with revised factor

With SF=1.2, the radial design force from the conservative 20 deg case is 43.46 N. For a 5 mm pin and 5 mm bearing contact, $p = 1.74 \text{ MPa}$. For a 6 mm pin and 6 mm bearing contact, $p = 1.21 \text{ MPa}$.

With SF=1.5, the radial design force from the conservative 20 deg case is 54.32 N. For a 5 mm pin and 5 mm bearing contact, $p = 2.17 \text{ MPa}$. For a 6 mm pin and 6 mm bearing contact, $p = 1.51 \text{ MPa}$.

8. Revised conclusion

Table 7. Revised design summary.

Result	Value
Single elevon area	0.04900 m^2
Dynamic pressure at 24 m/s	336.96 Pa
Conservative force on one elevon at 20 deg	36.22 N
Conservative hinge moment at 20 deg	1.434 Nm
Direct-drive design torque, SF=1.2 and $\eta=0.90$	$1.912 \text{ Nm} = 19.5 \text{ kg cm}$
Direct-drive design torque, SF=1.5 and $\eta=0.90$	$2.390 \text{ Nm} = 24.4 \text{ kg cm}$
Theoretical pin diameter at $\tau_{\text{allow}}=100 \text{ MPa}$, SF=1.5	4.96 mm
Practical minimum solid metal pin size	5 mm diameter x 30 mm length
Stiffer option if space allows	6 mm diameter x 30 mm length

With the revised safety factor, the practical minimum pin diameter drops from 7 mm to 5 mm for a metal pin with approximately 100 MPa allowable shear stress. A 6 mm pin remains preferable for stiffness, lower twist, and tolerance to misalignment or imperfect torque locking.

Important caution: reducing the safety factor to 1.2-1.5 leaves much less tolerance for underestimated hinge moment, higher-than-assumed flight speed, servo stall-torque marketing values, backlash, friction, and manufacturing defects. This is more suitable for prototype sizing than for a robust flight-critical design.